

JBHI Supplementary

Title: n-cylindrical Symbolic Response, a standalone and synergistic biomarker for Epilepsy diagnosis on EEG modality.

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Section I: Computational Complexity

The computational complexity of the nCSR feature vector along with the complete fused feature combination (i.e. FFT +DWT + nCSR) are evaluated in this section. Our proposed method used three features, namely f_1 , f_2 , and f_3 , fused in a way to obtain the optimized results. In our work, we used the f_1 feature as FFT, whose time complexity is, $O(k \log k)$, whereas the f_2 feature is for DWT, whose complexity is $O(k)$, and for our proposed f_3 feature i.e. nCSR(without overlap) (operations are done on a real symmetric covariance matrix), it is $O(k^2 \cdot \frac{n_{samples}}{f_s})$ thus, the final complexity is the sum of all the complexity of features which will be like:

$$f_1 + f_2 + f_3 \Rightarrow O(k \log k) + O(k) + O(k^2 \cdot \frac{n_{samples}}{f_s}).$$

Here f_s is the sampling frequency for our dataset, which is 125 Hz. Similarly, k , stands for number of trials, $n_{samples}$ is a total number of time samples in the EEG recording:

$$n_{Samples} = 3 \times 125 \times 60 = 22500$$

As per Big-O simplification principles in [1], the constants and lower-order terms are disregarded. Hence the highest is $O(k^2 \cdot \frac{n_{samples}}{f_s})$, so the final complexity will be $O(k^2 \cdot \frac{n_{samples}}{f_s})$

where $\frac{n_{samples}}{f_s} = \frac{22500}{125} = 180$. However, with 1 sample point overlap, the time complexity of the proposed nCSR feature will be $O(k^2 \cdot (n_{samples} - (f_s - 1)))$ where $n_{samples} - (f_s - 1) = 22376$.

Since in both cases i.e. overlap and non-overlap, the term $\frac{n_{samples}}{f_s}$ and $n_{samples} - (f_s - 1)$ remains constant throughout the dataset. So as per [1], the constant term does not impact asymptotic complexity in Big-O notation, as a result, the computational complexity of our nCSR features for both overlapping and non-overlapping cases reduces to $O(k^2)$. Hence, the final complexity of our proposed fused feature is $O(k^2)$.

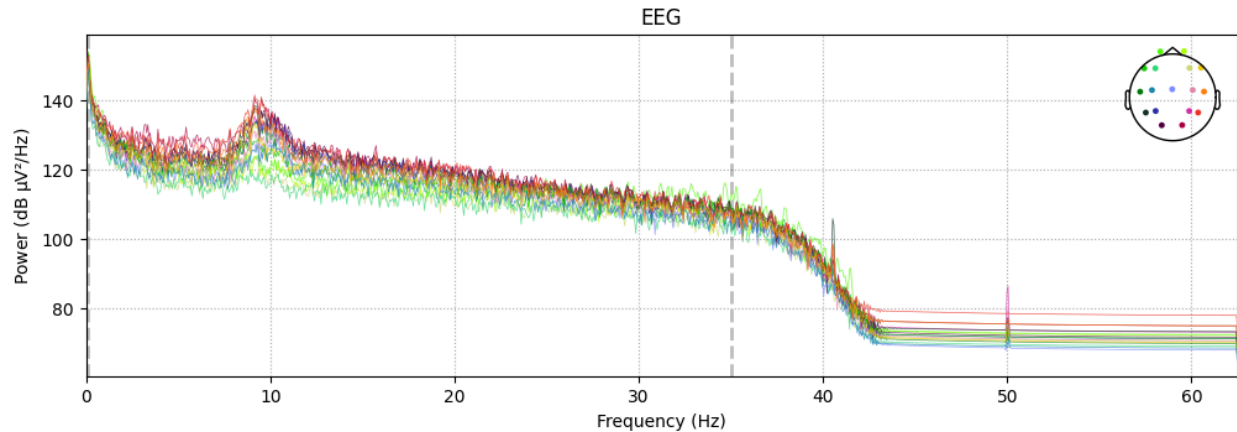
References:

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to algorithms. MIT press, 2022

Section II: Topo Plots for different EEG Datasets.

ICMR Epilepsy Dataset:

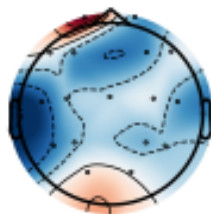
PSD Plot:



Discussion:

The topographic PSD distribution for the ICMR dataset reveals a pronounced concentration of spectral power over a focal region of the scalp, particularly in the lateral temporal area. This indicates that most interictal power is localized near the presumed seizure onset zone. Notably, one hemisphere (e.g., the left temporal lobe region around electrodes T3/T5) exhibits substantially higher power spectral density than the rest of the scalp, highlighting a clear focal hotspot. This focal power concentration suggests that seizures in the ICMR dataset are predominantly localized, across the cortex.

Average Window Topo Plot:

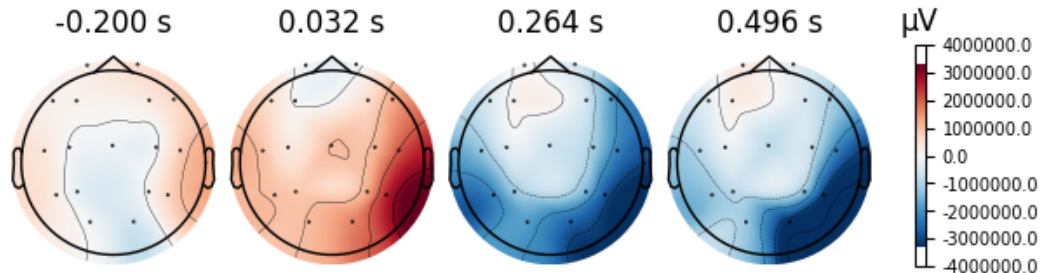


Discussion:

The average EEG activity topoplots similarly shows elevated amplitude confined to the same temporal region. Channels over the seizure focus consistently have higher average signal amplitudes during interictal windows, reflecting recurrent high-voltage epileptiform discharges in that area. Meanwhile, electrodes farther from the focus display markedly lower average activity, reinforcing the localized nature of the seizure events. This spatial pattern of average

activity aligns with the PSD findings, underscoring a single dominant region driving the epileptic activity.

Topo Plot for 5 sec:

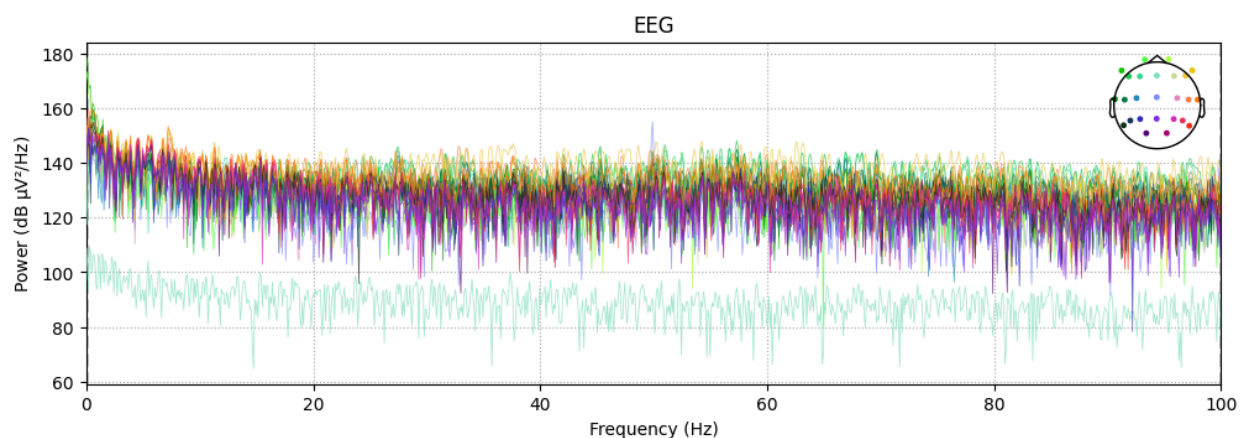


Discussion:

The ICMR dataset's topoplots consistently demonstrate a strong, lateralized focal seizure pattern. The PSD map's concentrated power and the average amplitude distribution both point to a well-defined epileptogenic zone (likely temporal lobe) on one side of the head. The 5-second interictal epileptic patient topomap confirms that epileptiform discharges start and largely stay confined to this region (at least in their early phase). This has clinical significance: a clearly localized scalp distribution often correlates with focal seizures that might be amenable to targeted therapies (e.g., resective surgery or localized neuromodulation).

BIOMED Epilepsy Dataset:

PSD Plot:

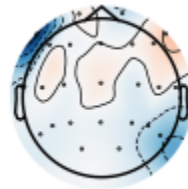


Discussion:

In the BIOMED dataset, the topographic PSD plot exhibits a more distributed pattern of power across the scalp compared to the ICMR data. Instead of a single sharply localized hotspot,

multiple regions show elevated spectral power. Thus indicating that the ictal activity is not confined to one isolated area.

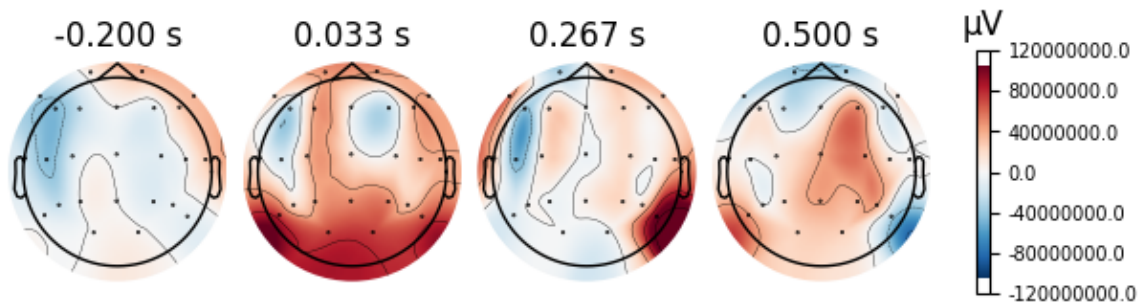
Average Window Topo Plot:



Discussion:

The average EEG amplitude topography reinforces this pattern of widespread involvement. During seizure windows, several electrode sites (spanning both left and right hemispheres) register above-average signal amplitudes. Rather than a single dominant channel, the BIOMED average average topoplot has multiple moderately “warm” regions, particularly over the temporal and frontal areas on both sides. This indicates that high-amplitude epileptiform activity is majorly present near the temporal and frontal region.

Topo Plot for 5 sec:

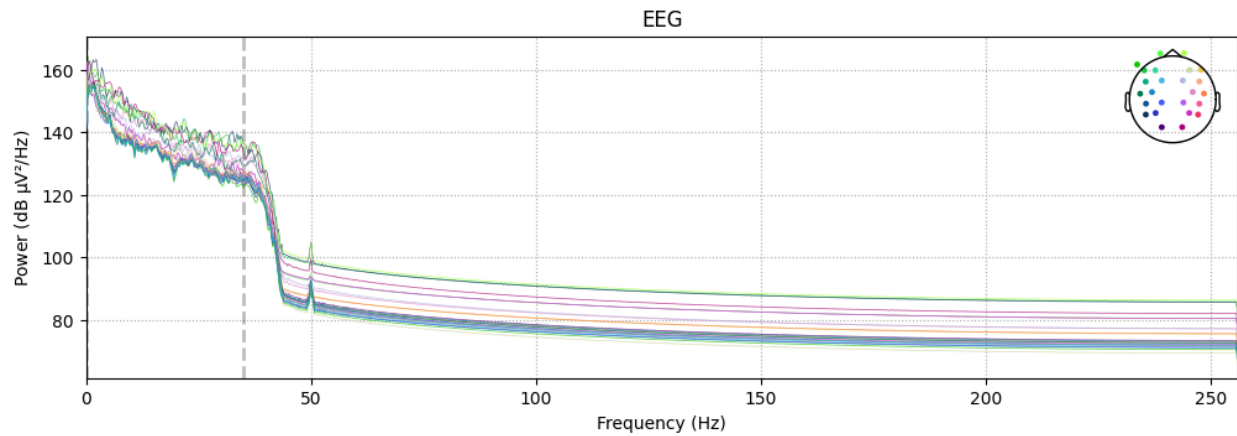


Discussion:

Looking at a 5-second seizure segment topomap, the spatial activation is seen covering a considerable portion of one hemisphere and often showing concurrent activation in the opposite hemisphere. In the topomap for 5 sec, an intense activity might be observed, for instance, in the right temporal area while simultaneously a milder increase appears in the homologous left temporal region – reflecting a propagation of seizure activity. Over the brief 5-second window, the high activity zone can be seen expanding or shifting, indicating dynamic spread of the seizure discharge. This example illustrates that the seizure activity in the BIOMED dataset tends to involve a network of regions rather than remaining stationary at one focus. The topomap’s coloration is generally more uniformly warm across broad areas, underscoring the extensive reach of the epileptic activity even in short time spans.

Siena Epilepsy Dataset:

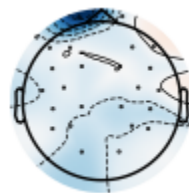
PSD Plot:



Discussion:

The Siena dataset's topographic PSD map also exhibits a concentrated region of high spectral power, consistent with a focal seizure origin. In the example topoplot from this dataset, the power is prominently elevated over a specific cortical area – for instance, a focus in the frontal region of one hemisphere (e.g., right frontal-central electrodes) shows the strongest PSD intensity. Surrounding areas have notably lower power, indicating a clear boundary around the seizure focus. While the overall pattern is localized similar to the ICMR dataset, the exact region of maximum power can vary (temporal in some patients, frontal in others), reflecting the diversity of focal epilepsy loci in the Siena cohort.

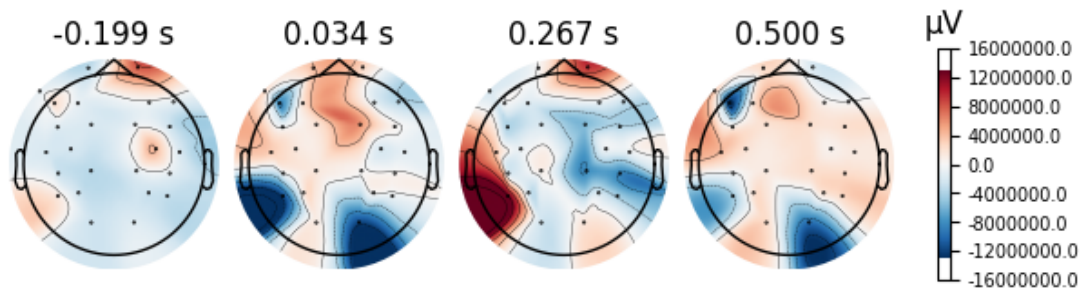
Average Window Topo Plot:



Discussion:

The average EEG activity topoplot for Siena similarly highlights a confined area of elevated amplitude. Electrodes covering the seizure region (in this case, the frontal lobe focus) register significantly higher average voltages during ictal periods compared to the rest of the scalp. This results in a distinct “hot” spot in the average amplitude map, whereas adjacent electrodes in the same hemisphere may show some moderate increase (indicating that the seizure's influence can spread locally).

Topo Plot for 5 sec:



Discussion:

In a representative 5-second segment topoplot, the evolution of the seizure activity in the Siena dataset can be observed. During a brief ictal window, the intense activity starts at the focal region (e.g., right frontal area) and may begin to propagate to nearby cortical areas on that same side. The topomap might show the main hotspot intensifying and slight extension of activity into adjacent electrode sites (for example, spreading from the frontal pole toward the central region), while the opposite hemisphere remains relatively inactive or only mildly affected. Throughout the 5-second snapshot, the highest amplitudes stay largely restricted to the originating hemisphere, indicating that any generalization is delayed or limited. This dynamic illustrates a contained propagation pattern typical of many focal seizures in adults – the discharge can travel within the affected hemisphere but does not immediately cover the entire brain.

Section III: nCSR Angular plot and Angular Density plot for Siena, BioMed, and CHB-MIT EEG Datasets.

Section III a) Siena Dataset:

Since both seizure and non-seizure classes originate from epilepsy patients, and therefore some degree of synchrony is present in both groups. However, the distinction lies in the strength and angular localization of synchrony. As shown in Figure 3, both seizure and non-seizure polar plots depicted by Fig. 3(a) and Fig. 3(b), exhibit clustering of nCSR angles, but the difference plot in Figure 3(c) highlights where the ictal condition dominates. Specifically, ictal EEGs show a markedly higher density of angular components in the 0° – 30° range, despite some synchrony also being present in the non-seizure class. This concentration reflects the heightened alignment of eigenvector orientations during seizures, which is consistent with pathological neuronal synchrony.

Figure 4 reaffirms this observation in linear density form. The seizure curve (blue) rises above the non-seizure curve (red) in the low-angle zone (0 – 0.5 rad $\approx 0^{\circ}$ – 30°), with the shaded region marking the dominance. While synchrony is indeed present in both groups, ictal EEGs consistently demonstrate stronger and more narrowly focused synchrony, making this subtle but important distinction clear.

Together, Figures 3(c) and 4 therefore provide complementary evidence that nCSR not only detects synchrony in epilepsy but also captures its intensified angular dominance during seizures, supporting its role as a computational biomarker.

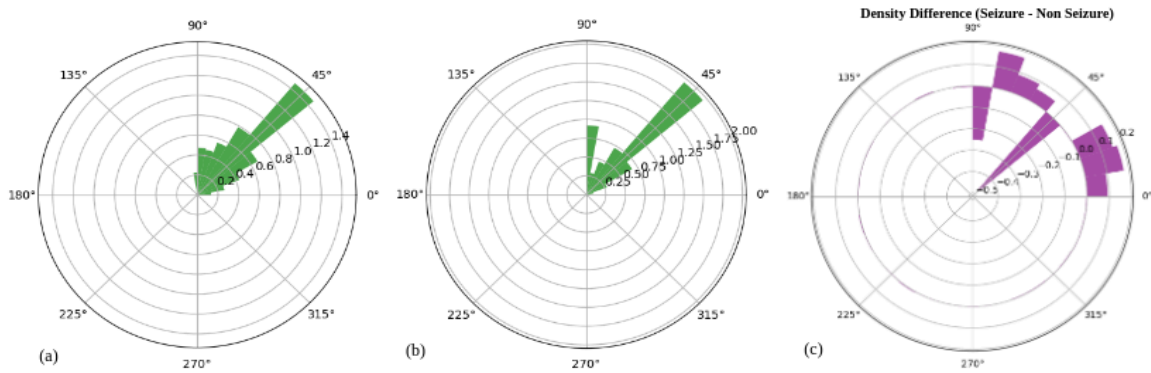


Figure 3: (a) nCSR angular plot for seizure set from Siena dataset, (b) nCSR angular plot for non-seizure set from Siena dataset, (c) nCSR angular plot difference between seizure set of Siena dataset and non-seizure set of Siena dataset.

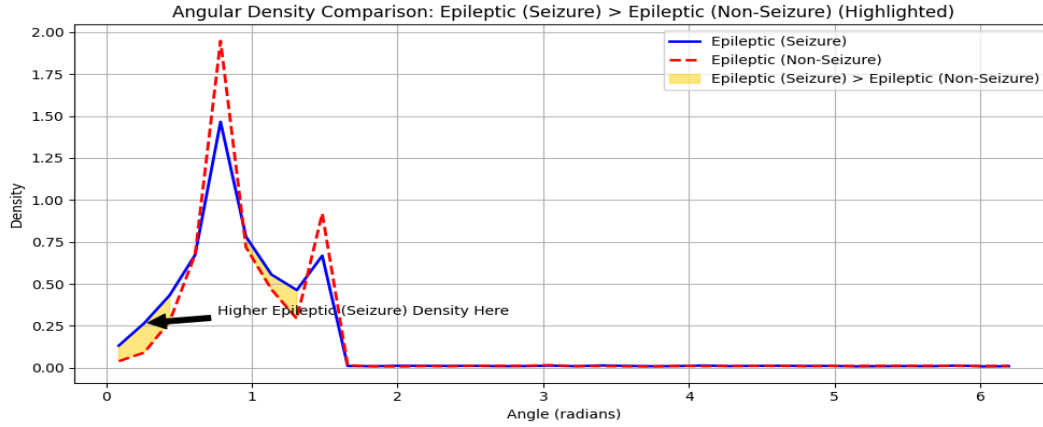


Figure 4: Angular Density Comparison for Siena Dataset: Epileptic (Seizure) > Epileptic (Non-Seizure)

Section III b) BioMed Dataset:

Figure 5(a) and 5(b) show the nCSR angular plots for the seizure and non-seizure classes, respectively. Both exhibit synchrony since they originate from epileptic patients. However, Figure 5(c), which isolates the angular density difference, reveals that the seizure class exhibits stronger clustering near the low-angle region (0° – 30°). The difference plot clearly highlights a positive density concentration around 0° – 15° , indicating that ictal EEGs contain more narrowly aligned eigenvector orientations compared to interictal EEGs. Figure 6 further strengthens this observation: the blue curve (seizure) consistently lies above the red curve (non-seizure) in the 0 – 0.5 radian ($\sim 0^\circ$ – 30°) range, showing a sharper angular concentration of synchrony during seizures. Beyond this region, the density difference diminishes, suggesting that the most discriminative separation between seizure and non-seizure states lies in the low-angle synchrony domain.

Thus, even though synchrony is present in both conditions due to the epileptic cohort, the seizure state demonstrates a more dominant and localized synchrony pattern, which the nCSR biomarker captures effectively.

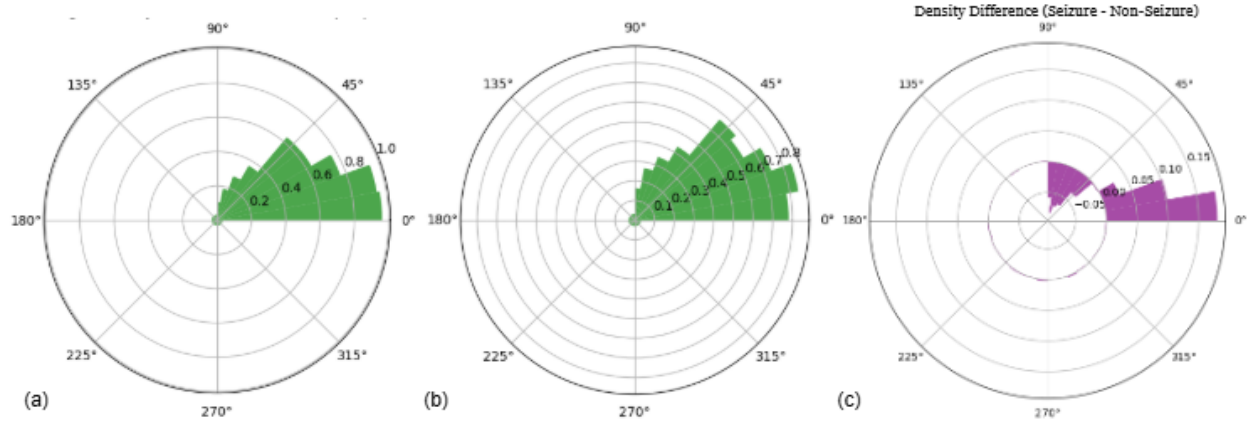


Figure 5: (a) nCSR angular plot for seizure set from BioMed dataset, (b) nCSR angular plot for non-seizure set from BioMed dataset, (c) nCSR angular plot difference between seizure set of BioMed dataset and non-seizure set of BioMed dataset.

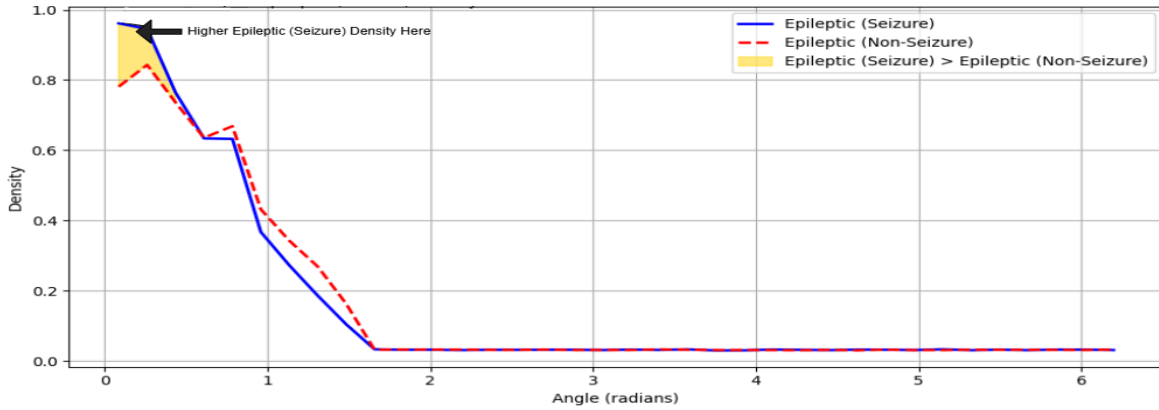


Figure 6: Angular Density Comparison for BioMed dataset: Epileptic (Seizure) > Epileptic (Non-Seizure)

Section III c) CHB-MIT Dataset:

Figure 7(a) shows the nCSR angular distribution for the seizure class, while Figure 7(b) presents the corresponding distribution for the non-seizure class. Both plots reveal synchrony, which is expected as all recordings come from epileptic patients. However, the differential plot in Figure 7(c) highlights a clear distinction i.e., seizure epochs exhibit a stronger density concentration in the low-angle band (0° – 30°), while non-seizure epochs show relatively weaker clustering in this range. Additionally, Figure 8 further validates this observation through the angular density curves. The seizure curve (blue) lies above the non-seizure curve (red) within ~ 0 to 0.5 radians ($\sim 0^{\circ}$ – 30°), with the yellow band marking regions of higher seizure density. Beyond 30° , the two curves largely overlap, indicating reduced discriminative information.

Thus, while synchrony is present in both classes (given the epileptic population), the seizure state demonstrates sharper, low-angle clustering (0° – 30°), representing heightened neuronal synchrony. This pattern is consistent with the findings from the Siena and BioMed datasets, reinforcing that nCSR captures ictal-specific synchrony shifts in a compact and interpretable manner.

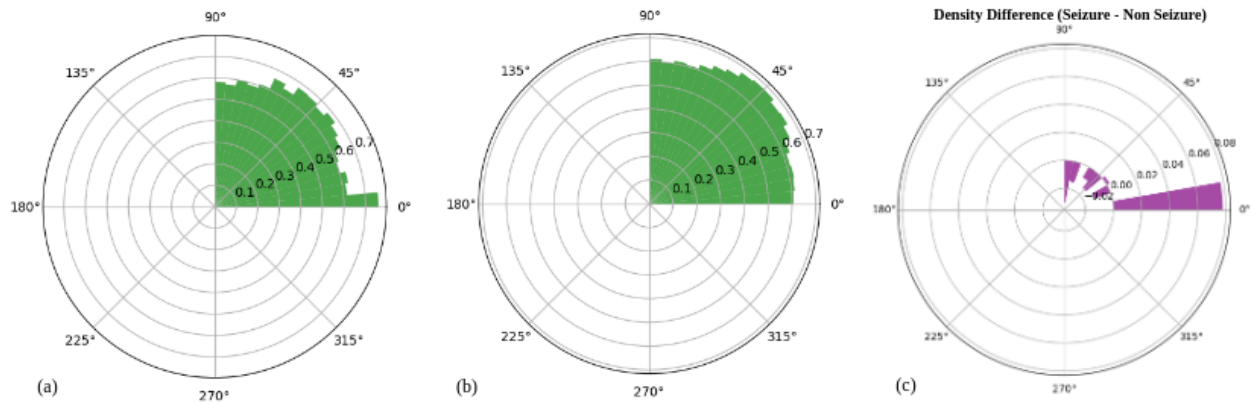


Figure 7: (a) nCSR angular plot for seizure set from CHB-MIT dataset, (b) nCSR angular plot for non-seizure set from CHB-MIT dataset, (c) nCSR angular plot difference between seizure set of CHB-MIT dataset and non-seizure set of CHB-MIT dataset.

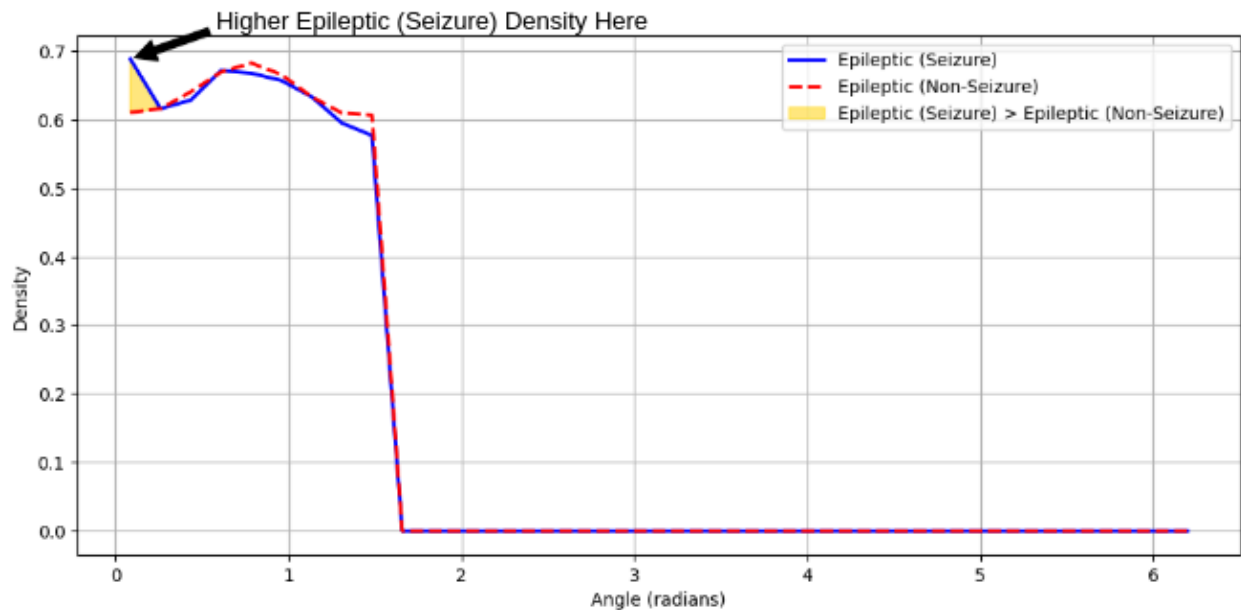


Figure 8: Angular Density Comparison for Siena Dataset: Epileptic (Seizure) > Epileptic (Non-Seizure)

Section IV: Hyperparameter Tuning.

For training, 12th Gen Intel(R) Core(TM) i7-12700K 64-Core Processor and 128 GB of RAM, equipped with an Nvidia RTX 3090ti (24 GB) GPU with driver version: 470.199.02 and CUDA version: 11.4 were used respectively. The models use Keras and Tensorflow backend with version '2.11.0'. To avoid overfitting in all datasets, Random Forest used 150 estimators, with a maximum depth of 15 and maximum features equivalent to 200. It used a minimum sample leaf of 50 whereas a minimum sample split of 24.

In this study, we conducted k-fold cross-validation on a binary class dataset (healthy control and interictal, or ictal and interictal) to assess the performance of the proposed feature and models. The dataset was divided into k-folds, with each fold serving as the validation set once, while the remainder was used for training. Optimal hyperparameters for each model were identified using grid search in each fold. The average accuracy across all folds was used to estimate the overall performance of the model. For our models, we received the best performance for $k = 5$ for all the datasets.

Section V: Statistical Analysis.

To validate our nCSR biomarker statistically, we performed independent two-sample Welch's t-tests [1] for each dataset, comparing nCSR feature values between healthy and epileptic groups for ICMR Epilepsy dataset, whereas for Siena dataset, CHB-MIT dataset and BIOMED datasets respectively we performed two-sample welch's t-tests [1] for seizure and non seizure groups. Here, in our analysis we selected Welch's t-test because it does not assume equal variances between groups, making it suitable for our heterogeneous EEG data [1]. A threshold of $p < 0.05$ was used for statistical significance.

1. ICMR: Welch's t-test showed a highly significant difference ($t \approx 5.50$, $p < 0.0001$), indicating the mean nCSR feature value was significantly higher in the healthy group.
2. SIENA: Welch's t-test revealed a significant difference ($t \approx -6.26$, $p < 0.0001$), with the mean nCSR value lower in the seizure group.
3. CHB-MIT: Welch's t-test demonstrated an extremely significant difference ($t \approx -21.81$, $p < 0.0001$), again with a lower mean in the seizure group.
4. BIOMED: Welch's t-test also showed a substantial difference ($t \approx -11.61$, $p < 0.0001$), with lower mean values for the seizure subjects.
5. BONN: Welch's t-test demonstrated an extremely significant difference.
 - a. $t \approx -4.31$, $p < 0.001$, for Set A vs Set E, where Set A is healthy (eyes open) and Set E is Ictal (seizure) class.
 - b. $t \approx -8.21$, $p < 0.001$, for Set B vs Set E, where Set B is healthy (eyes closed) and Set E is Ictal (seizure) class.

References:

1. Welch, B.L., 1947. The generalization of 'STUDENT'S' problem when several different population variances are involved. Biometrika, 34(1-2), pp.28-35.

Section VI: Study Objective Table.**Table I.** Study Objective over each database.

S. No	Dataset	Dataset Type	Study Objective
1.	ICMR Epilepsy Dataset	In-Situ	Epilepsy Diagnosis
2.	University of Bonn Dataset	Public	Epilepsy Diagnosis and Seizure Detection
3.	BIOMED Dataset	Challenge Dataset	Seizure Detection
4.	Siena Dataset	Public	Seizure Detection
5.	CHB-MIT Dataset	Public	Seizure Detection